Sarvesh Gharat

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Visiting Researcher at MBZUAI (2026) — Multi-Agent Systems for Scientific Discovery

Education

IIT Bombay — Mumbai, In
PhD in Artificial Intelligence and Data Science — *July 2022 – May 2027*

Vishwakarma Institute of Information Technology — Pune, In
BTech in Electronics and Telecommunication — *Aug. 2018 – May 2022*

Research Interests

My research focuses on online learning and sequential decision-making, with applications to reliable and resource-efficient LLM reasoning. My current interests also include continual learning and multi-agent systems for scientific discovery.

Experience

- **Summer Research Intern** — May 2025 – July 2025
Adobe Research — *Host: Soumyabrata Pal, Ramasuri Narayanam*
  - Improved LLM reasoning under fixed compute constraints by developing a novel method to inject auxiliary information during sequential test-time scaling.
  - Designed and implemented an efficient, on-the-fly example selection strategy using embedding similarity, leading to more relevant in-context examples during inference.
  - Focused on designing inference-time techniques that enhance model responses without changing any underlying parameters.

Patent Filed: Gharat, S., Pal, S., Narayanam. R., Guided Sequential Test Time Scaling using additional hints from exemplars

- **Student Researcher** — Oct 2024 – March 2025
Google DeepMind — *Host: Aparna Taneja, Milind Tambe*
  - Evaluated AI-targeted intervention scheduling in a large-scale maternal health program using data from thousands of beneficiaries.
  - Designed comparative analyses against matched control groups to measure effects on engagement, maternal health knowledge, and postnatal care behaviors.
  - Developed robust analyses for noisy and variable survey responses, enabling reliable detection of intervention effects.

- **Visiting Researcher** — Aug 2021 – June 2022
Tata Institute of Fundamental Research — *Host: Prof. Shriganesh Prabhu*
  - Designed optimization methods to estimate the refractive index of optically thin materials using Terahertz Time Domain Spectroscopy.
  - Implemented and benchmarked algorithms including Dual Annealing, SHGO, and Newton-Raphson to improve convergence stability.
  - Built a user-friendly Python library to support experimental analysis and make the refractive index estimation pipeline accessible for both researchers and educators.

Peer-Reviewed Publications

- **Gharat, S.**, The Argmax Gap in Human Chess Move Prediction. To appear in Transactions on Machine Learning Research (TMLR)
- **Gharat, S.**, Karamchandani, N., and Nair, J., Cost-Aware Best LLM Identification using Dueling Feedback. To appear at NeurIPS 2026
- Singh, V., Ghosal, S., **Gharat, S.**, Pal, S., Narayanam. R., Manocha. D, ThinkRetrieve: Retrieval-Augmented Reasoning Traces for Test-Time Scaling. To appear in Findings of the Association for Computational Linguistics: EMNLP 2026

- **Gharat, S.**, Karamchandani, N. and Nair, J., 2026, May. Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models. In Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (pp. 3600-3602).
- Dasgupta, A.*, **Gharat, S.***, Madhiwalla, N., Hegde, A., Tambe, M., Taneja, A., 2025, October. AI-Targeted Calls Drive Measurable Improvements in Maternal Health, In Proc. of European Conference on Artificial Intelligence (pp. 5200-5207).
- **Gharat, S.**, Yadav, A., Karamchandani, N. and Nair, J., 2025, April. Representative arm identification: A fixed confidence approach to identify cluster representatives. In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
- Suresh, R., Bishnoi, H., Kuklin, A.V., Parikh, A., Molokeev, M., Harinarayanan, R., **Gharat, S.** and Hiba, P., 2024. Revolutionizing physics: a comprehensive survey of machine learning applications. Frontiers in Physics, 12, p.1322162.
- Bhatta, G., **Gharat, S.**, Borthakur, A. and Kumar, A., 2024. Gamma-ray blazar classification using machine learning with advanced weight initialization and self-supervised learning techniques. Monthly Notices of the Royal Astronomical Society, 528(1), pp.976-986.
- Mistry, P., Prasad, A., Maity, M., Pathak, K., **Gharat, S.**, Lekkas, G., Bhattarai, S., Kumar, D., Lissauer, J.J., Twicken, J.D. and Soubkiou, A., 2024. VaTEST III: Validation of eight potential super-earths from TESS data. Publications of the Astronomical Society of Australia, 41, p.e030.
- **Gharat, S.**, Borthakur, A. and Bhatta, G., 2024. Estimation of redshift and associated uncertainty of Fermi/LAT extragalactic sources with Deep Learning. Monthly Notices of the Royal Astronomical Society, 527(3), pp.6198-6210.
- **Gharat, S.**, Bose, B., Borthakur, A. and Mazumder, R., 2023. An image processing approach to identify solar plages observed at 393.37 nm by the Kodaikanal solar observatory. RAS Techniques and Instruments, 2(1), pp.393-397.
- Mistry, P., Pathak, K., Prasad, A., Lekkas, G., Bhattarai, S., **Gharat, S.**, Maity, M., Kumar, D., Collins, K.A., Schwarz, R.P. and Mann, C.R., 2023. VaTEST. II. Statistical Validation of 11 TESS-detected Exoplanets Orbiting K-type Stars. The Astronomical Journal, 166(1), p.9.
- Pininti, V.R., Bhatta, G., Paul, S., Kumar, A., Rajgor, A., Barnwal, R., and **Gharat, S.**, 2023. Exploring short-term optical variability of blazars using TESS. Monthly Notices of the Royal Astronomical Society, 518(1), pp.1459-1471.
- **Gharat, S.** and Dandawate, Y., 2022. Galaxy classification: a deep learning approach for classifying Sloan Digital Sky Survey images. Monthly Notices of the Royal Astronomical Society, 511(4), pp.5120-5124.
- Kulkarni, J.S., Cengiz, K. and **Gharat, S.**, 2021, March. Compact C-slot microstrip-fed planar antenna for wireless devices. In 2021 International Conference on Emerging Smart Computing and Informatics (ESCI) (pp. 651-654). IEEE.

** Equal contribution.*

Workshop Papers

- Babu, AG., Rastogi, M., Anand, A., Chen, C., **Gharat, S.**, MACA: Tracing and Correcting Bias Evolution in Multi-Agent LLM Deliberation. NeurIPS SocialAgent 2026
- **Gharat, S.**, Komiyama, J., PRISM: Problem Discovery via Structural Motifs in Knowledge Graphs. ICML AI for Science Workshop 2026.
- **Gharat, S.**, Borthakur, A. and Bhatta, G., Gamma Ray AGNs: Estimating Redshifts and Blazar Classification using Neural Networks with smart initialization and self-supervised learning. NeurIPS Machine Learning and the Physical Sciences Workshop 2023.

Manuscripts Under Review

- **Gharat, S.**, Karamchandani, N. and Nair, J., Learning What Users Prefer: Online LLM Selection with Feedback Acquisition Costs
- **Gharat, S.**, Hebbalaguppe, R., Not All Thoughts Are Worth Thinking: Reliability-Aware Test-Time Scaling
- **Gharat, S.**, Komiyama, J., SGHA: Evidence-Grounded Research Problem Discovery with Local Language Models
- **Gharat, S.**, Chavan, U., Karamchandani, N. and Nair, J., Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning
- Kumar, A., **Gharat, S.**, Beyond One-Shot Decisions: Progressive Morphological Reasoning in Galaxy Classification

Invited Talks & Presentations

- **Invited Talks**
  - Which Model to Train? A Budgeted Bandit Approach to Adaptive Fine-Tuning
    — Cohere for AI Reinforcement Learning Group (2026)
  - *Cost-Aware Best Arm Identification via Dueling Feedback with Applications to Large Language Models*
    — Cohere for AI Research Connections (2025)
  - *Representative arm identification: A fixed confidence approach to identify cluster representatives*
    — Cohere for AI Reinforcement Learning Group (2025)
  - *Estimating Redshifts of AGNs using Neural Networks*
    — IAU-IAA Astroinformatics Seminar (2024)
- **Poster Presentation:**
  - ACM India ARCS 2026
  - Google Research Week 2024
  - IndoML Graduate Forum 2024
  - ACM Pingala Interactions in Computing 2024

Academic Service & Highlights

- **Reviewer / Program Committee**
  - ICLR 2027
  - IJCAI AI for Social Good (2025, 2026) - Awarded Bronze Tier reviewer distinction in 2026
  - KDD AI for Sciences 2026
  - AAMAS (2025, 2027)
  - IEEE Transactions on Information Theory
- **Research Grants and Support**
  - Awarded *Thinking Lab Research Grant* for the project on "An Online Learning Approach for Adaptive Fine-tuning"
  - Selected for the *Cohere Labs Catalyst Grant Program* for a proposal on online learning
  - Inaugural cohort grantee of the *Adaption Labs Research Grant*, for research on continual learning, online learning, and adaptive AI systems
  - State Bank of India (SBI) funded project (2024–2026) on *Dynamic LLM Optimization and Fine-Tuning via Dueling Bandits*, exploring adaptive model selection under resource constraints

- **Selected Programs**
  - AAMAS Doctoral Consortium 2026
  - ACM India ARCS 2026
  - Google Research Week (2024, 2025)
  - IndoML Graduate Forum 2024
  - ACM CODS-COMAD PhD Clinic 2024
  - ACM Pingala Interactions in Computing 2024
- **Kaggle Expert**
  - LLM Prompt Recovery Challenge: Silver Medal (Rank 79/2175)
  - Santa 2025 Challenge: Bronze Medal (Rank 228/3357)

Teaching Experience

- **CS 747:** Foundations of Intelligent and Learning Agents **(Best TA Award)**
- **CS 663:** Digital Image Processing
- **EE 325:** Probability and Random Processes
- **DS 203:** Programming for Data Science

Mentorship Experience

- Algoverse AI Research Program (May 2026 – Ongoing)
- Krittika Astronomy Summer Project, IIT Bombay 2026
- Summer of Sciences, IIT Bombay 2023

Technical Skills

- **Programming**: Python, LaTeX, Apps Script
- **Libraries**: PyTorch, TensorFlow, NumPy, pandas, vLLM, Transformers
- **Tools**: Jupyter Notebook, Git, Docker, Google Cloud Platform, VS Code